

A finite Hurwitz-kernel cocycle for periods and L -functions of meromorphic modular forms on $\mathrm{SL}_2(\mathbb{Z})$

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Abstract

We give one finite, regulator-free contour functional — a two-segment cocycle integral with an explicit Hurwitz-zeta T -kernel — that realizes both the period polynomials and the L -functions of modular forms on $\mathrm{SL}_2(\mathbb{Z})$ whose poles may lie anywhere: inside the fundamental domain and at the cusp. From a finite base point $\tau_0 \in \mathbb{H}$ one builds a cocycle whose cohomology class is independent of τ_0 . Removing its T -component by a coboundary — possible if and only if the constant Fourier coefficient $c_f(0)$ vanishes — yields the parabolic representative, whose S -value is the period polynomial. The same kernel, with the period monomial τ^ℓ replaced by τ^{s-1} , gives the completed L -function on all of $M_k^!$, constant term allowed, with the classical poles of Λ ; for forms with interior poles it is determined by the contour up to the residues crossed. The construction introduces no new objects: the periods are Brown’s quasi-periods in closed form [1], the L -values are those of Bringmann–Fricke–Kent [2]. The contribution is the explicit all- s kernel and the single finite realization. All formulae are checked numerically to 50 digits [12].

1 A finite L -integral

Definition 1.1 (S - and T -segments). *Let $\tau_0 \in \mathbb{H}$ and $f \in F_k$, and let $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ generate $\Gamma = \mathrm{SL}_2(\mathbb{Z})$. The S - and T -segments at τ_0 are smooth paths in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$,*

$$\gamma_{\tau_0}^S : -1/\tau_0 \longrightarrow \tau_0, \quad \gamma_{\tau_0}^T : \tau_0 - 1 \longrightarrow \tau_0, \quad (1)$$

running from $S^{-1}\tau_0 = -1/\tau_0$ and $T^{-1}\tau_0 = \tau_0 - 1$ to τ_0 .

The pole-avoidance is inert for $f \in M_k^!$: with no interior poles, $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty) = \mathbb{H}$ and the segments are unobstructed. Interior poles — and the homotopy classes of contours they create — become relevant only for meromorphic f , in the final section.

Definition 1.2 (Hurwitz T -kernel). *The Hurwitz T -kernel is, with $\zeta(s, x) = \sum_{m \geq 0} (m + x)^{-s}$ the Hurwitz zeta function (meromorphic in s , holomorphic in $x \in \mathbb{H}$),*

$$\tilde{k}_T(\tau, s) := \zeta(1 - s, \tau + 1) - e^{i\pi(s-1)} \zeta(s - (k - 1), \tau + 1). \quad (2)$$

Definition 1.3 (L -integral). *Let $f \in F_k$, $\tau_0 \in \mathbb{H}$, and $s \in \mathbb{C}$, and let $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ be the S - and T -segments at τ_0 , as in Definition 1.1. The L -integral of f is*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := i^{-s} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right]. \quad (3)$$

Its τ_0 - and path-dependence is dealt with below, in the main body.

Definition 1.4 (*L-function*). *The L-function of $f \in F_k$ is*

$$L(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T). \quad (4)$$

Likewise; once that dependence is settled below, we drop the arguments and write simply $L^*(f, s)$ and $L(f, s)$.

Lemma 1.5 (Finiteness). *For $f \in F_k$, both integrals in Definition 1.3 are finite for every $s \in \mathbb{C}$, every $\tau_0 \in \mathbb{H}$, and every choice of paths $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$.*

Proposition 1.6 (Basepoint and path independence). *For $f \in M_k^!$ the value $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is independent of $\tau_0 \in \mathbb{H}$ and of the paths: with no interior poles, $\mathbb{H}$ is simply connected, so all admissible contours are homotopic. We write $L^*(f, s)$ for the common value and $L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s)$.*

Remark 1.7 (Classical Mellin recovery). *For a holomorphic cusp form $f \in S_k$, the cusp limit $\tau_0 \rightarrow i\infty$ collapses the S -segment onto $[0, i\infty]$ while the T -segment vanishes exponentially, so*

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s) = \int_0^\infty t^{s-1} f(it) dt, \quad (5)$$

the classical completed L-integral.

2 The L-function

Proposition 2.1 (Cusp unfolding). *For $n > 0$ and $\nu \in \mathbb{C}$,*

$$\int_i^{i+1} e^{2\pi i n u} \zeta(\nu, u) du = i^{1-\nu} (2\pi n)^{\nu-1} \Gamma(1-\nu, 2\pi n). \quad (6)$$

Theorem 2.2 (*L-function for weakly holomorphic forms*). *For $f \in M_k^!$, at $\tau_0 = i$,*

$$L^*(f, s) = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right] = i^k L^*(f, k-s). \quad (7)$$

Definition 2.3 (Period cocycle). *Let $V_n := \mathbb{C}[X, Y]_n$ with the action $(P|_\gamma)(X, Y) := P(aX + bY, cX + dY)$, and $\omega_f := f(\tau) (X - \tau Y)^n d\tau$. For $f \in F_k$ and $\tau_0 \in \mathbb{H}$, the period cocycle is*

$$C_\gamma^f(\tau_0) := (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} \omega_f \in V_n, \quad \gamma \in \Gamma, \quad (8)$$

satisfying $C_{gh}^f = C_g^f|_h + C_h^f$.

Lemma 2.4 (Parabolic representatives exist iff $c_f(0) = 0$). *With $\delta_T P := P|_T - P$ on V_n , one has $[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} c_f(0)$, so $C_T^f \in \text{im } \delta_T$ if and only if $c_f(0) = 0$ — the principal part of f never obstructs.*

Definition 2.5 (Period polynomial). *For $f \in S_k^!$ (so $c_f(0) = 0$), the period polynomial is the S -value of the parabolic representative,*

$$r_f := C_S^f(\tau_0) - (P|_S - P) \in V_n, \quad (9)$$

with $P \in V_n$ chosen so that $\delta_T P = C_T^f$ (Lemma 2.4).

Proposition 2.6 (Period coefficients are special L -values). *For $f \in S_k^!$ and $\ell \in \{1, \dots, k-3\}$,*

$$[r_f]_{X^{n-\ell}Y^\ell} = (2\pi i)^{n+1}(-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1). \quad (10)$$

Corollary 2.7 (Period coefficients are quasi-period coordinates). *For every weight with $\dim S_k = 1$ — $k \in \{12, 16, 18, 20, 22, 26\}$ — and a weak cusp form $f \in S_k^!$, the coefficients of r_f (Proposition 2.6) are the quasi-period coordinates $\omega^\pm(f)$ in the sense of [1]: ω^+ from ℓ even, ω^- from ℓ odd. At $k = 12$ they reproduce Brown's published $\eta^\pm(\widehat{\Delta})$; at the remaining five weights they furnish the quasi-periods anew [12].*

Remark 2.8 (Period-determinant cross-check). *The extracted periods and quasi-periods satisfy, numerically, a Legendre-type relation: the determinant $\omega^+\eta^- - \omega^-\eta^+$ lies in $\mathbb{Q}^\times \cdot (2\pi i)^{k-1}$, equal to $-(k-2)!(2\pi i)^{k-1}$ at $k = 12$ in Brown's normalization (the weight- k analog of $\omega_1\eta_2 - \omega_2\eta_1 = 2\pi i$). This is a global consistency check on the extraction, claiming nothing new [12].*

2.1 Why a finite contour

Remark 2.9. *For a weak cusp form $f \in S_k^!$ the classical Mellin integral $\int_0^{i\infty} f(\tau)(\tau/i)^s \frac{d\tau}{\tau}$ does not converge — f grows at the cusp — and the deformed-Mellin construction of [2] supplies a finite value through a non-holomorphic regularization there. The functional of Definition 1.3 reaches the same value with no regularization: written as its two segments, each diverges as $\tau_0 \rightarrow i\infty$, with opposite signs, and the two cancel at every finite τ_0 , so $L^*(f, s)$ is finite throughout $\mathbb{H}$ although the Mellin contour itself has no limit. By Proposition 1.6 the value is independent of τ_0 and of the paths; BFK's regularized value is its evaluation at $\tau_0 = i$ (Theorem 2.2), and its integer- s values are Brown's quasi-periods (Corollary 2.7). Two independent finite computations — Brown's cohomological periods and the BFK L -values, obtained without reference to one another — meet at this one contour, which fixes it as the canonical object. Finiteness and τ_0 -independence are the analytic shadow of the parabolic representative [1, 5, 6].*

3 Meromorphic modular forms

Definition 3.1 (Polylog projection). *Let $f \in F_k$ have interior poles at the points τ_p of a set $\mathcal{P}_f$ of Γ -inequivalent representatives, and write $\text{Li}_s(z) = \sum_{n>0} z^n n^{-s}$. Since $\text{Li}_{1-m}(e^{2\pi i(\tau-\tau_p)})$ has a pole of order m at τ_p , the principal part of f there is matched by a unique polylog projection*

$$\widehat{P}_{\tau_p} f(\tau) := \sum_{m>0} r_{f, \tau_p}^*(m) \text{Li}_{1-m}(e^{2\pi i(\tau-\tau_p)}), \quad (11)$$

its coefficients $r_{f, \tau_p}^(m)$ being the principal-part data of f at τ_p . The projection $\widehat{P} := \sum_{\text{im } \tau_p \geq 1} \widehat{P}_{\tau_p}$ removes the principal part of every pole on or above the line $\text{im } \tau = 1$.*

Definition 3.2 (Regularized cusp coefficients). *With $\widehat{P}$ as in Definition 3.1,*

$$c_{Rf}(n) := [(I - \widehat{P})f]_{q^n}. \quad (12)$$

Definition 3.3 (The space $F(\infty)$). *$F(\infty)$ is the space of meromorphic g on $\mathbb{H}$ with $g(\tau) = g(\tau+1)$, finitely many poles in any compact subset of the strip $\{\text{Im } \tau > 0, |\text{Re } \tau| \leq \frac{1}{2}\}$, and $g(\tau) = O(e^{C \text{Im } \tau})$ as $\text{Im } \tau \rightarrow \infty$ for some C . Each such g has a Fourier expansion $g = \sum_n c_g(n) q^n$.*

Then $F_k \subset F(\infty)$, and $(I - \widehat{P})f \in F(\infty)$ — the polylog tails are T -periodic but not modular, so the projection leaves F_k .

Lemma 3.4 (q -series–integral commutation). *Let $g \in F(\infty)$ have no pole on or above the line $\text{Im } \tau = 1$ (e.g. $g = (I - \widehat{P})f$). Then $g = \sum_n c_g(n)q^n$ converges along the T -segment at $\tau_0 = i$ and integrates term by term:*

$$\int_{\gamma_i^T} g(\tau) \widetilde{k}_T(\tau, s) d\tau = \sum_n c_g(n) \int_{i-1}^i e^{2\pi i n \tau} \widetilde{k}_T(\tau, s) d\tau. \quad (13)$$

A pole at $\text{Im } \tau_p \geq 1$ defeats this — there the coefficients grow at least like $n^{m-1}e^{2\pi n \text{Im } \tau_p}$ [3, Lem. 2.6].

Definition 3.5 (Polylog integral). *For $N \geq 0$ and $z \in \mathbb{H}$, the polylog integral is the L -integral of a single tail $\text{Li}_{-N}(e^{2\pi i(\tau-z)})$ along the T -segment at $\tau_0 = i$ (where γ^S degenerates, since $-1/i = i$):*

$$I_N(s, z) := i^{-s} \int_{i-1}^i \text{Li}_{-N}(e^{2\pi i(\tau-z)}) \widetilde{k}_T(\tau, s) d\tau. \quad (14)$$

Lemma 3.6 (Closed form of the polylog integral). *Write $\mathcal{G}(s, m) := \frac{\Gamma(s, 2\pi m)}{(2\pi m)^s} + i^k \frac{\Gamma(k-s, 2\pi m)}{(2\pi m)^{k-s}}$, the building block of Theorem 2.2 continued to $m < 0$. For the projected poles $\text{Im } z > 1$, where $|e(\tau - z)| \geq 1$ on the contour, polylog inversion $\text{Li}_{-N}(w) = (-1)^{N+1} \text{Li}_{-N}(1/w)$ resums the integral into the convergent series*

$$I_N(s, z) = \delta_{N,0} \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} \sum_{n \geq 1} n^N e^{2\pi i n z} \mathcal{G}(s, -n). \quad (15)$$

Theorem 3.7 (L -function of a meromorphic modular form). *For $f \in F_k$ with no pole on the edge $\text{Im } \tau = 1$, at $\tau_0 = i$,*

$$L^*(f, s) = -c_{Rf}(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_{Rf}(n) \mathcal{G}(s, n) + \sum_{\text{Im } \tau_p > 1} \sum_{m > 0} r_{f, \tau_p}^*(m) I_{m-1}(s, \tau_p), \quad (16)$$

$\mathcal{G}(s, n) = \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}$; the regularized cusp sum (Definition 3.2, Theorem 2.2) plus one polylog integral (Lemma 3.6) per interior pole above the edge. Poles on the edge enter by the principal value of Lemma 3.8.

Lemma 3.8 (L -function, a pole on the edge). *Routing the T -path just below versus just above a pole τ_p of $f \in F_k$ at height $\text{Im } \tau_p = 1$ gives finite values differing by the residue,*

$$L_{\text{below}}^*(f, s) - L_{\text{above}}^*(f, s) = 2\pi i i^{-s} \text{Res}_{\tau=\tau_p} [f(\tau) \widetilde{k}_T(\tau, s)]. \quad (17)$$

We define the edge value as the principal value $L^* = \frac{1}{2}(L_{\text{above}}^* + L_{\text{below}}^*) = L_{\text{below}}^* - \pi i i^{-s} \text{Res}_{\tau=\tau_p} [f \widetilde{k}_T]$. At the elliptic point $\tau_p = i$ the pole lies on the S -segment instead; its principal value, reached as $\tau_0 \rightarrow i$, is finite and basepoint-independent.

Corollary 3.9 (Wall-crossing). *For $f \in F_k$, $L^*(f, s)$ is constant in τ_0 within each chamber of $\mathbb{H} \setminus f^{-1}(\infty)$; crossing a wall, where the contour sweeps past an interior pole τ_p , changes it by*

$$\Delta L^*(f, s) = 2\pi i i^{-s} \text{Res}_{\tau=\tau_p} [f(\tau) \widetilde{k}_T(\tau, s)] \quad (18)$$

when the T -segment crosses τ_p (and by the same with τ^{s-1} in place of $\widetilde{k}_T$ when the S -segment does). Thus L^* is fixed by the contour up to the residues it crosses — the meromorphic counterpart of Proposition 1.6, recovering its constancy on $M_k^!$.

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